

 VIT[®] BHOPAL www.vitbhopal.ac.in	Differential and Difference Equations												Course Type		LT		
Course Code:		MAT 2001												Credits		3	
Prerequisite:		Calculus and Laplace Transforms															
Course Objectives: • This course is designed to give a comprehensive coverage at an introductory level to the subject of ordinary differential equations and difference equations. • Matrix methods and eigenvalue problems are integrated in the course. • Sufficient emphasis is laid on mathematical modelling and analysis of simple engineering problems.																	
Course Outcomes (CO): Students will be able to																	
CO1: Apply the concept of eigenvalue and Eigen vectors in developing engineering models and their desirable results. KL2, KL3																	
CO2: Apply and analyze the concept of FS and FT in signal and systems. KL3 , KL4																	
CO3: Apply and analyze the concept of Z Transform for the Discrete Functions. KL2, KL3																	
CO4: Understand Difference Equation and solution of Difference Equation using Z Transform. KL3, KL4																	
Correlation of COs with POs																	
CO \ PO	CKL	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	
PKL		3	5	6	5	6	3	3	3	NA	M	3	M	3	3		
CO1	3	3	3	3	2	2			2	2	2	2	2				
CO2	4	3	3	3	3	2			2	2	2	2	2				
CO3	3	3	3	3	2	3	2	2		2	2	2	2				
CO4	4	3	3	2	3	3				2	2	2	2				
CO		Topics to be discussed														Lecture	
CO1		Matrix methods to linear differential equations Eigen value problem - Eigen values and eigen vectors – properties of eigen values and eigen vectors – diagonalization using similarity and orthogonal transformations – solution of system of linear first order differential equations with constant coefficients														6	
CO2		Fourier series Introduction to Fourier series - Euler formulas for the Fourier coefficients - Dirichlet's conditions – Fourier convergence Theorem (Statement) – Fourier series over a shifted interval - Half range Fourier series – Parseval's identity – Frequency and amplitude spectra of a function - computation of harmonics														6	
CO2		Fourier Transform Fourier Integral Theorem (statement only), Fourier Transform of a function, Fourier Sine and Cosine Integral Theorem (statement only), Fourier Cosine & Sine Transforms of elementary functions. Properties of Fourier Transform: Linearity, Shifting, Change of scale. Examples. Fourier Transform of Derivatives. Examples. Convolution Theorem (statement only), Inverse of Fourier Transform, Examples. Application of Fourier Transform in Real Life.														6	
CO3		Z transform Definition of Z-transform - Z-transform of discrete sequences and functions – Relation between Z transform and Laplace transform – Definition of inverse Z transform – Inverse Z transform by partial fraction and convolution methods.														5	
CO4		Difference Equations Introduction to Difference equation – order and linear - Fibonacci sequence - solution of														5	

	linear difference equations with constant coefficients - complementary functions - particular integrals by the method of undetermined coefficients - solution of simple difference equations using Z-transforms	
	Guest Lectures by experts on contemporary topics	2
	Total Lecture: (1 Lecture = 1.5 Hrs.)	30
Text Books: 1. Advanced Engineering Mathematics by Erwin Kreyszig, 10 th Edition, Wiley India Pvt. Ltd., New Delhi, 2015. 2. Higher Engineering Mathematics by B. S. Grewal, 43 rd Edition, Khanna Publishers, India, 2014.		
Reference books: 1. Advanced Engineering Mathematics by Peter V. O' Neil, 7 th edition, Cengage Learning, 2012. 2. Elementary differential equations and boundary value problems by William E. Boyce, Richard C. DiPrima, 10 th edition, John Wiley & Sons, Inc., 2012 3. Advanced Engineering Mathematics by Michael E. Greenberg, 2 nd edition, Pearson, 2006		
<i>Recommendation by the Board of Studies on</i>		22-November- 2018
<i>Approval by Academic council on:</i>		27-June-2018
<i>Compiled by:</i>		Dr. Manisha Jain, Dr. Reena Jain, Dr. Anant Kant Shukla